

Nhan, Phan Ky (Matteo)

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EDUCATION

Ho Chi Minh University of Technology (HCMUT) – VNU
B.Sc. in Computer Science | Expected GPA: 3.65/4.00

Ho Chi Minh City, Vietnam
Oct 2024 – Expected 2028

Luong Van Chanh High School for the Gifted
Specialized in Mathematics

Phu Yen, Vietnam
Sep 2021 – July 2024

RESEARCH INTERESTS

My research focuses on bridging 3D/4D neural representations (e.g., 3D Gaussian Splatting) with physically-grounded scene understanding and efficient ML systems. I aim to develop scalable, trajectory-respecting models that natively capture real-world dynamic mechanics while leveraging low-level system optimization to ensure high-performance, amortized simulation.

INDEPENDENT RESEARCH PROJECTS

High-Performance GPT-2 Inference Engine in Pure C & CUDA

Jan 2026 – Present

- Engineered a single-model, forward-inference-only GPT-2-124M engine from scratch. Achieved true M=1 decode speeds of 511.7 tokens/s, reaching 76–83% of `llama.cpp` F16 matched-precision performance.
- Executed a rigorously gated optimization ladder featuring tiled shared-memory GEMM, online softmax flash attention, and KV caching with true M=1 GEMV. Validated metrics against an HF fp16 oracle and measured CUDA-core ceilings via custom microbenchmarks.
- Documented honest negative results: weight-only INT8 yielded 0% speedup due to launch bounds at small batch sizes, and fused LayerNorm+matmul was 2.9% slower than isolated kernels.

Label-Efficiency & Synthetic Data Study for Vietnamese OCR

Sep 2025 – Dec 2025

- Conducted a pre-registered, ablation-controlled label-efficiency study on Vietnamese scene-text OCR (evaluated on the VinText dataset) by building a custom generator and a three-axis NFD scorer.
- Showed that at a scarce 10% label budget, synthetic data reduces recognition-only CER from 16.51% to 13.73%.
- Demonstrated through control arms that the gain primarily stems from supplying sequence-level length signals to repair premature decoder termination, rather than through aggressive domain-realistic pixel degradation.

RESEARCH EXPERIENCE

ML4U Lab, HCMUT AI Tech Lab
AI Undergraduate Researcher Supervised by Prof. Duc Dung Nguyen

Ho Chi Minh City, Vietnam
April 2026 – Now

- **Research Project:** NeuInteract

- Investigating amortized, controllable contact deformation on 3D Gaussian Splatting (physics-grounded, native-on-density, trajectory-respecting); ran a pre-registered FEM validation campaign (FEniCSx mixed- (u, p) Taylor–Hood, cross-solver verified) that characterized contact-induced surface bulge as a linear incompressible-confinement effect ($D_{bulge} \approx 3a$, bonded-base gated), and conducted adversarial prior-art occupancy analysis to scope the contribution. ([Project is ongoing, targeted for CVPR 2027 submission](#))

Ticklab

AI Research Engineer

Ho Chi Minh City, Vietnam

May 2025 – April 2026

• Research Project: Real-time ECG Classification on Edge Devices

- Researched and optimized an energy-efficient Spiking Neural Network (SNN) pipeline for real-time ECG classification on edge devices, overcoming severe class imbalance (mode collapse) and significantly improving the macro F1-score for minority arrhythmia classes.

TECHNICAL SKILLS

- **Languages & Core Computing:** C/C++, CUDA, Python, Linux/WSL, Bash, Make, Git.
- **GPU Profiling & Optimization:** Nsight Compute (ncu), cuobjdump/SASS, cuBLAS, CUDA events, Roofline analysis.
- **Deep Learning & 3D Vision:** PyTorch, llama.cpp, Hugging Face (Diffusers, Accelerate), Nerfstudio, OpenCV.

AWARDS & ACHIEVEMENTS

- **Datathon 2026: THE GRIDBREAKER:** Led the team in designing Entity-Relationship Diagrams (ERDs) and architecting a robust machine learning pipeline, placing in the top 5% of the Kaggle submission leaderboard, but we got knocked out caused our shortage of business foundation (2026)
- **Active Competitor — HCMC AI Challenge 2025:** Gained practical hands-on experience by prototyping and evaluating machine learning models on real-world datasets within a strict competitive timeframe. (2025)
- **Academic Merit Scholarship:** Awarded for maintaining outstanding academic performance across multiple semesters. (Fall 2024 [3.8], Spring 2026 [3.9])
- **Second Prize – Provincial Academic Competition in Mathematics:** Awarded for achieving top ranks among high school students in the province-wide examination. (2022)

LEADERSHIP & COMMUNITY INVOLVEMENT

• AI Competition Team Lead

Self-organized | 2025 – Now

- Led a 3-member AI competition team, designing practical training roadmaps covering deep learning fundamentals and system implementation to prepare for hackathons.

• Independent AI & Tech Content Sharing

2024 – Now

- Distilled academic concepts into accessible study materials shared across technical social media communities (HCMUT forums on Facebook). Selected topics: [Naive Bayes Classification](#), [Linear Programming](#), [Computer Vision Journey](#), [Variational Wasserstein Clustering](#)...